

Response to Reviewers

Manuscript: Trust in Autonomous Human-Robot Collaboration: Effects of Responsive Interaction Policies

We thank both reviewers for their careful reading of the manuscript and for comments that have strengthened our paper. We address each point below in the order it was raised. For completed revisions, we indicate the specific changes made and their location in the revised manuscript.

Reviewer 1

Comment 1: Bundled Manipulation (Responsiveness)

“The authors state on Page 14 that ‘responsiveness... reflects a bundled interaction profile rather than a single isolated manipulation.’ This point should be emphasised much earlier in the manuscript, as it is essential for understanding the scope and limits of the study’s contributions.”

We agree that this concept needed to be fleshed out before the results rather than within them. In the revised manuscript we added a paragraph at the end of the **Interaction Policies** subsection in Methods that names the dimensions along which the two conditions differ simultaneously (dialogue content, affective framing, check-in style, frequency of proactive guidance, session duration, and robot response latency), explains why these features are functionally interdependent, and frames the responsive condition explicitly as a bundled interaction profile rather than a single-factor manipulation. The original passage in the Descriptive Results section has been shortened to a brief cross-reference to the Methods discussion, rather than re-introducing the concept as though it were new information.

Comment 2: Sample Size — Toning Down Claims

“The Abstract states that ‘Sensitivity analyses indicated a pattern consistent with affective and experiential components...’ This phrasing implies that the authors have identified differential sensitivity among trust subcomponents, but such a conclusion would require a substantially larger sample and a validated multi-factor measurement model... Similarly, the claim that the ‘findings suggest that trust in autonomous human-robot interaction emerges from process-level interaction dynamics...’ is overstated for a small, exploratory pilot study.”

We agree with both points. Two changes were made:

Abstract, sentence 1: The phrase “affective and experiential components of trust” implied sub-scale-level differentiation, which is not warranted at $N = 24$ without a validated multidimensional measurement model. We have revised the abstract to refer explicitly to the two instruments (TI-HRC and TPS-HRI), to state that the responsive advantage held convergently across both, and to present the instrument-level difference descriptively rather than as a tested effect: “A tentative pattern in which experienced trust (TI-HRC) appeared more sensitive to communication breakdown

than perceived trust (TPS-HRI) is reported descriptively; the small sample size and exploratory design preclude formal tests of differential sensitivity across trust dimensions.”

Abstract, concluding sentence: “These findings suggest that trust... emerges from process-level interaction dynamics” has been revised to: *“These preliminary findings are consistent with the interpretation that trust in autonomous human-robot interaction is shaped by process-level interaction dynamics.”* The causal framing has been softened throughout.

A matching revision was made to the parallel passage in the Discussion (Section 4, paragraph 3), where the instrument-level comparison is now presented with an explicit caveat about sample size and the absence of a validated multidimensional model.

In addition, the Introduction (within the Theoretical Rationale and Study Predictions subsection) now explicitly introduces the two trust instruments and the facets they capture, framing the predicted advantage as convergent across both rather than as a basis for contrasting trust dimensions—reinforcing, at the point of prediction, that any instrument-level difference is descriptive and not formally tested.

Comment 3: Novelty — Communication Breakdown in Prior Literature

“The idea that communication breakdown harms trust is not new. Prior HRI research has shown that communication breakdown undermines trust (for example, Salem et al., 2015; Robinette et al., 2016). Thus, the claim in the Introduction that such breakdowns are ‘under-studied’ seems overstated.”

We appreciate the reviewer raising this and pulled up these papers for a read and consequently have added citations to both papers, as well as to a recent (2025) systematic review of HRI trust research (Gratšjova et al., 2025). However, we would respectfully note that the cited papers address a related but distinct phenomenon: Salem et al. (2015) examine the effects of robot behavioural errors (task mistakes) on cooperation and perceived reliability, while Robinette et al. (2016) demonstrate overtrust in emergency evacuation following observable navigation failures. Neither paper specifically addresses ASR-driven speech-channel failure — the form of breakdown central to our study.

We have revised the Introduction to acknowledge this prior work explicitly while making a more precise gap claim. The revision makes the following argument: the underrepresentation of ASR-driven communication failure is not a blind spot in the literature but a structural consequence of prior methodological choices. Under Wizard-of-Oz paradigms spoken-language understanding is performed by a human operator who to some extent absorbs speech-channel failures invisibly. Communication viability therefore does not vary as an observable experimental variable in WoZ research. This class of failure only becomes empirically tractable in fully autonomous systems, and large language model-driven dialogue management has only recently made such systems feasible for “in-the-wild” collaborative interaction. The revised paragraph cites Salem et al. (2015), Robinette et al. (2016), and Gratšjova et al. (2025) in this context.

Comment 4: Acronym Management

“NARS (Negative Attitudes Toward Robots Scale) is introduced twice on Page 10 within a single paragraph... ASR is used without ever providing the full term.”

Both issues have been corrected.

Reviewer 2

Comment 1: Results Subsection Titles

“The sub-section titles in Results are named after methods/models rather than content-driven conclusion-oriented headings.”

All seven Results subsection headings have been revised to be conclusion-driven. The changes are as follows:

Original	Revised
Descriptive Analysis: Eligible Sample	Responsive Interaction Was Associated with Higher Trust Independent of Task Accuracy
Hierarchical Models	Policy Effects on Trust Were Confirmed Across Frequentist and Bayesian Models
Primary Analysis: Eligible Sample (n = 24)	Responsive Policy Produced Credible Increases in Both Trust Measures (n = 24)
Sensitivity Analysis: Full Sample (n = 29)	Including Sessions with Communication Failure Attenuated but Did Not Eliminate the Policy Effect (n = 29)
Mechanism Analysis: Communication Breakdown (n=29)	Communication Breakdown Moderated the Trust Advantage of Responsive Interaction (n = 29)
Task performance	Trust Differences Were Not Attributable to Differential Task Accuracy
Individual differences and correlational patterns	Trust Correlated with Interaction Dynamics Rather Than Individual Differences or Task Outcomes

Comment 2: Method Flowchart

“It is strongly recommended to add a graphical flowchart of the experimental workflow to simplify presentation and improve readability.”

A graphical flowchart of the experimental procedure and subsequent dialogue coding and classification procedure is now included in the Methods section (Figure 2).

Comment 3: Theoretical Framework and Research Hypotheses

“It is suggested to add a dedicated Theoretical Foundation and Research Hypotheses section, and explicitly propose progressive hypotheses to highlight the core research logic.”

We have added a **“Theoretical Rationale and Study Predictions”** subsection at the end of the Introduction. Given that this study was conducted as an exploratory pilot, and that we believe transparency about the a priori vs. exploratory status of analyses is important, we have framed the section around *directional predictions* rather than formal confirmatory hypotheses. This framing reflects the study’s design and preserves the distinction between what was predicted in advance and what emerged from data collection.

Essentially there were only two a priori directional predictions: (P1) that responsive interaction would be associated with higher post-interaction trust compared to the neutral, reactive interaction; and (P2) that responsive interaction would benefit collaborative task performance. The section also identifies two theoretically motivated covariates: a) NARS (baseline negative attitudes toward robots) and b) Need for Cognition, and explicitly flags the communication viability analyses as post-hoc and exploratory, arising from real-world ASR constraints that we did not anticipate rather than a designed manipulation. We explicitly tried to screen out participants where spoken English was not fluent during the sign-up screening but some participants seem to have overestimated their language proficiency and/or underestimated the difficulty of the task. Though we did not anticipate the degree to which language would breakdown in this way, we believe that it is an important feature of the study that speaks to the real-world viability of speech-based interaction and also a direction for future research.

Comment 4: Introduction — Differentiation from Prior Work

“The motivation and research gaps are not sufficiently refined in Introduction. The differences between this work and existing HRI trust studies should be further summarized and emphasized.”

We agree that the Introduction, while addressing the WoZ/ASR gap in some detail (see also Response to Reviewer 1, Comment 3), did not provide a direct, consolidated statement of what distinguishes this study from the broader HRI trust literature. In the revised manuscript, we have added a dedicated paragraph at the end of the study introduction (immediately following the

statement of objectives) that explicitly contrasts this work with prior HRI trust research on three specific axes:

1. **Full autonomy.** All spoken-language understanding, dialogue management, and response selection were performed autonomously, with no live, human operator involvement. Participants were exposed to genuine ASR errors, response latencies, and misinterpretations rather than idealized scripted exchanges.
2. **Speech-channel viability as an empirical variable.** Natural variability in ASR success across sessions made communication viability an observable empirical variable—one that is structurally inaccessible in WoZ paradigms where operator-mediated comprehension absorbs speech-channel failures before they affect the interaction.
3. **Policy-level responsiveness under uncontrolled conditions.** The affect-responsive condition was implemented as an integrated system, coordinating dialogue management, real-time affect inference, and task-state tracking, rather than as a scripted surface behaviour or isolated manipulation.

This paragraph complements the revised novelty argument already introduced in the context of prior ASR/failure literature (Response to Comment 3, Reviewer 1), and together these revisions we hope provide a more complete and direct account of the study’s contribution relative to prior work.

We have also strengthened the surrounding motivation. The Introduction now classifies the system’s autonomy explicitly under an established levels-of-robot-autonomy framework (Beer et al., 2014) and situates it against a systematic review of autonomy in socially assistive robotics (Maure and Bruno, 2025), clarifying that prior work on responsive and affect-adaptive robots has relied on Wizard-of-Oz or pre-programmed, rule-based control rather than full autonomy. The measurement literature is likewise grounded with reference to a systematic review of trust assessment in HRI (Campagna and Rehm, 2025).

Comment 5: Limitations — Sample Size, Randomization, Session Exclusions

“The pilot study limitation is vaguely stated; the small sample size, non-full randomization and exclusion of non-viable interaction sessions should be explicitly acknowledged and discussed more thoroughly in Limitations.”

We agree that the prior Limitations section addressed these issues too briefly. Three additions have been made:

Sample size. The opening paragraph has been expanded to acknowledge not only limited statistical power but also the unequal cell sizes that resulted from attrition ($n = 10$ responsive, $n = 14$ neutral in the eligible sample). We clarify that the study should be treated as generating directional evidence rather than definitive estimates, and emphasized our explicit note that replication with larger, balanced samples is required.

Non-full randomization. A new paragraph explicitly describes what happened: participants were initially randomized at sign-up via Qualtrics, but attrition among scheduled participants led to replacement participants being assigned to the next available session slot without re-randomization, producing a semi-random final allocation. We report that randomization checks on all measured baseline covariates (NARS, NFC, age, gender, prior robot experience) revealed no systematic group differences (and include those tables in the report), while acknowledging that this cannot rule out residual confounding from unmeasured characteristics.

Session exclusion and selection bias. A new paragraph acknowledges that the exclusion of non-viable sessions, while methodologically justified, constitutes a source of potential upward bias in the primary effect estimates: excluded participants were disproportionately those with lower ASR recognition rates, and if their profiles differed from retained participants in ways associated with trust outcomes, the eligible-sample effects could be inflated. We note that full-sample sensitivity analyses and mechanism-focused comparisons of excluded sessions partially address this risk, while acknowledging that they do not fully resolve it. We also recommend that future studies implement more robust ASR-compatibility pre-screening or a more flexible ASR system that can handle accent/language differences to reduce attrition from communication failure.

Comment 6: Discussion — Theoretical Interpretation

“The discussion section merely restates results rather than deeply interpreting underlying mechanism and theoretical implications. Please strengthen the theoretical interpretation of why responsive interaction enhances trust and why communication breakdown moderates such effect.”

We agree that the prior Discussion restated findings without adequately connecting the dots or exploring the underlying mechanisms. On re-reading, we also felt the prose was more clinical and less readable than it should be, and we have rephrased much of it to convey the findings’ real-world meaning more directly—without altering any claims or their associated caveats. In the same pass, we moved the Limitations and Future Directions material into subsections of the Discussion rather than standalone sections, so they no longer interrupt its flow as they are part of the discussion. Among these changes, the Discussion now includes two paragraphs that explicitly address the points raised:

Why responsive interaction enhances trust. We have added a dedicated mechanistic paragraph after the statement of the main result. It identifies three complementary mechanisms: (1) *social presence* — proactive engagement cues signal social attentiveness, and perceived attentiveness is associated with affective trust independent of task outcome; (2) *social exchange and reciprocity* — unsolicited assistance constitutes a relational investment, and participants may extend reciprocal trust to a robot that proactively supports them without being prompted; (3) *mutual grounding* — proactive dialogue facilitates the shared understanding of task state and affective engagement necessary for participants to feel both understood and effectively coordi-

nated with their robotic partner, supporting both competence-based and rapport-based trust. These mechanisms are explicitly linked back to P1 from the Theoretical Rationale section.

Why communication breakdown moderates the effect. The prior “viability window” paragraph described the pattern but not the mechanism. We have expanded it to consider *why* the responsive advantage specifically attenuates under communication failure: the affect-responsive policy depends on an intact dialogue channel to perceive user state, infer affect, and generate contextually appropriate responses. When the speech channel degrades, the robot’s affect inference breaks down, severing the feedback loop through which responsiveness is expressed. In this degraded state, proactive responses generated from misrecognized input may be experienced as incoherent, potentially amplifying the negative effects of failure rather than offsetting them. We also note the practical implication: deployed systems using affect-responsive policies should incorporate more robust communication monitoring and graceful degradation strategies. The revised paragraph ties this back to the exploratory communication breakdown analyses and to P2 from the Theoretical Rationale.

Comment 7: Appendix — Condense Technical Details

“Some methodological details and repetitive descriptions in Appendix can be condensed; raw technical implementation contents that do not serve core research logic should be properly streamlined.”

Appendix A has been revised to remove or substantially condense content that does not serve the core research logic:

- **Software Architecture:** The explicit SDK dependency list (Deepgram SDK, FFmpeg, Flask and Flask-SocketIO, DuckDB) has been removed. The revised section notes that full dependency details are available in the project repository and retains only the high-level architectural statement.
- **Dialogue Management:** The sections on LangChain integration and prompt structure have been merged and condensed. Specific generation parameters (temperature, memory buffer details, JSON logging of conversation histories) have been removed, retaining only information relevant to understanding the system’s dialogue architecture.
- **Prompt Structure:** The detailed description of JSON field injection has been condensed to two sentences describing the general approach.
- **Data Collection and Logging:** DuckDB-specific implementation details have been removed; the section now briefly notes the logging structure without specifying the underlying technology.
- **Inter-process Communication:** The Flask REST endpoint description has been removed; communication between the web task interface and the dialogue agent is now noted in a single sentence within the Collaborative Tasks section.

- **Interaction Dynamics and Control Policy:** Minor condensation to eliminate redundancy with the main Methods section.

The retained content covers the hardware platform overview, dialogue management and LLM architecture, speech processing pipeline, affect inference, task design, and control policy – the elements directly relevant to evaluating the study’s methods and replicating its design.